

Lab: Data Visualization Using Statistical Charts

Estimated reading time: 30 minutes

In this lab, you will learn to create a column chart for the given data set. You will also be able to design a pie chart, line chart, and waterfall chart for the given data set.

Software Used in this Lab

The instruction videos in this course use the full Excel Desktop version, as this has all the available product features. However, for the hands-on labs, we will use the free 'Excel for the web' version as this is available to everyone.

Although you can use the Excel Desktop software if you have access to this version, [it is recommended that you use Excel for the web for the hands-on labs](#), as the lab instructions specifically refer to this version. There are some small differences in the interface and available features.

Data set Used in this Lab

The dataset used in this lab is curated from the following source: <https://www.kaggle.com/datasets/varunhlakant/mobile-phone-prices> under the [CC0 Public Domain license](#).

Acknowledgment and thanks also go to <https://wpak.in>, who were generous enough to share the data publicly for free.

We are using a modified subset of that data set for the lab. To follow the lab instructions successfully, please use the data set provided in this lab rather than the data set from the original source.

Learning Objectives:

- Design a column chart to compare data
- Create a pie chart and define its use for the percentage analysis
- Design a line chart to identify trends
- Create a waterfall chart to identify changes in data

Welcome to the lab: Data Visualization using Statistical Charts!

Exercise: Design Various Statistical Charts

Let's consider you need to analyze [monthly sales data](#). (To download the given data set in this lab, right click on the given hyperlink and select Open link in new tab) for various mobile phone brands, such as iPhone, Samsung, and OPPO, for the year 2022. You want to use various data visualization techniques to represent the given data statistically. To solve this problem, you can open Excel for the web and perform the following tasks.

Task 1: Design a Column Chart

In this task, you'll learn to design the column chart for the given data set. The column chart compares the monthly sales data for three mobile phone brands: iPhone, Samsung, and OPPO. To do so, follow the steps given below.

1. Copy and paste the data set into an Excel for the web worksheet.

The screenshot shows the Excel for the web interface with the following data set:

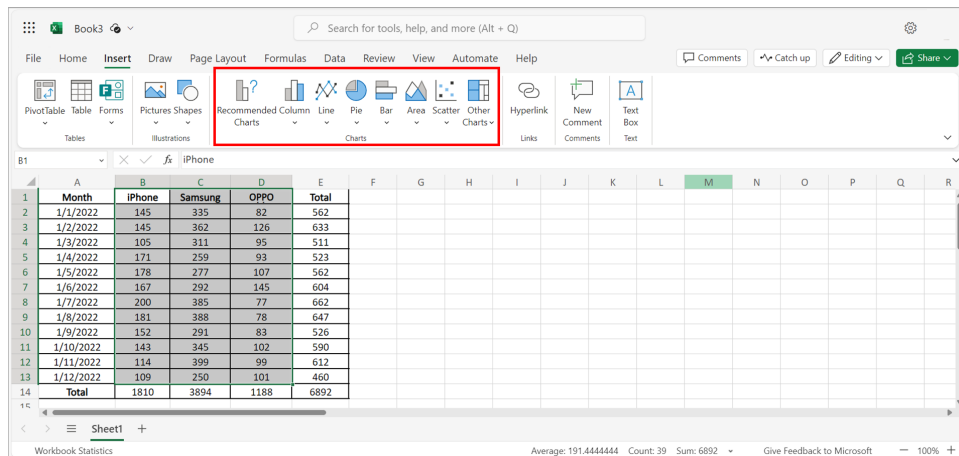
Month	iPhone	Samsung	OPPO	Total
1/1/2022	145	335	82	562
1/2/2022	145	362	126	633
1/3/2022	105	311	95	511
1/4/2022	171	259	93	523
1/5/2022	178	277	107	562
1/6/2022	167	292	145	604
1/7/2022	200	385	77	662
1/8/2022	181	388	78	647
1/9/2022	152	291	83	526
1/10/2022	143	345	102	590
1/11/2022	114	399	99	612
1/12/2022	109	250	101	460
Total	1810	3894	1188	6892

2. Select the data for which you want to design the column chart and then select the **Insert** tab.

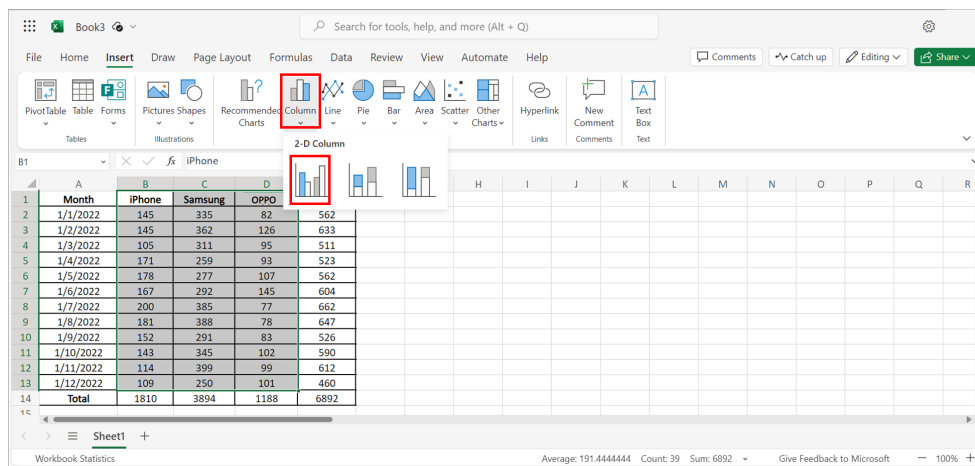
The screenshot shows the Excel for the web interface with the **Insert** tab selected. The data set is the same as in the previous screenshot, but the **Insert** tab is highlighted in the ribbon, and the data range B1:E13 is selected.

Month	iPhone	Samsung	OPPO	Total
1/1/2022	145	335	82	562
1/2/2022	145	362	126	633
1/3/2022	105	311	95	511
1/4/2022	171	259	93	523
1/5/2022	178	277	107	562
1/6/2022	167	292	145	604
1/7/2022	200	385	77	662
1/8/2022	181	388	78	647
1/9/2022	152	291	83	526
1/10/2022	143	345	102	590
1/11/2022	114	399	99	612
1/12/2022	109	250	101	460
Total	1810	3894	1188	6892

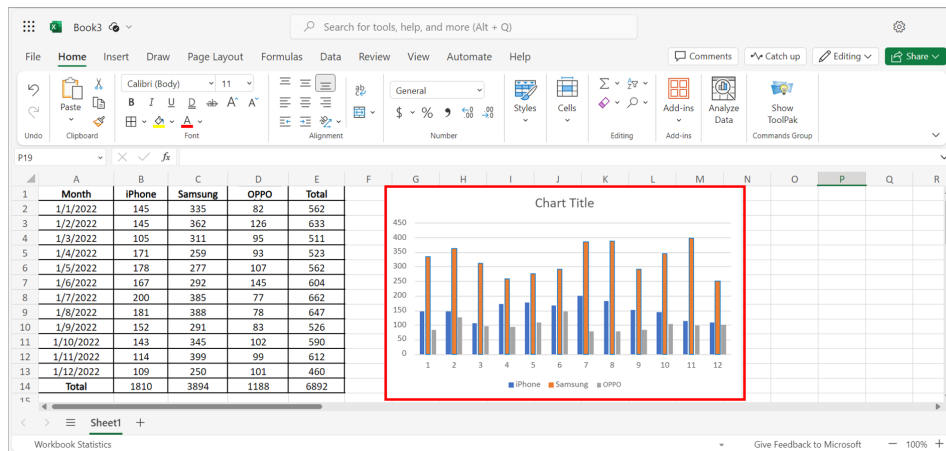
3. You can see various chart options.



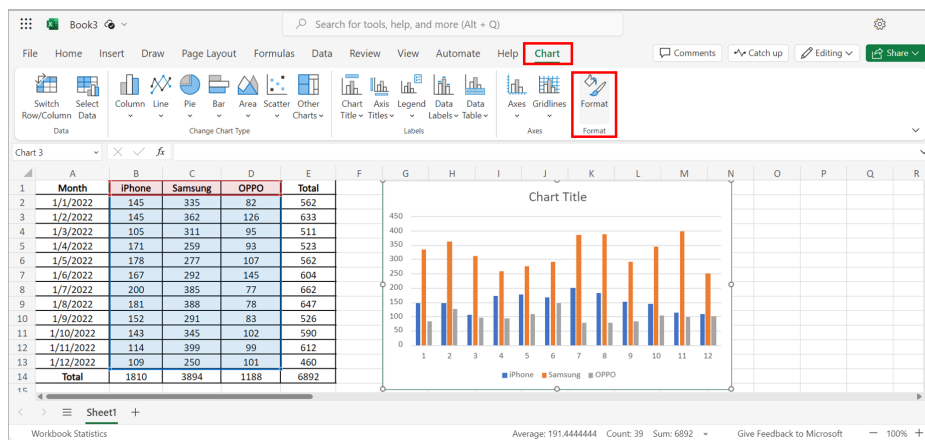
4. Select **Column** chart. You'll see possible column charts. Choose the appropriate chart. In this case, it is **2-D Column**.



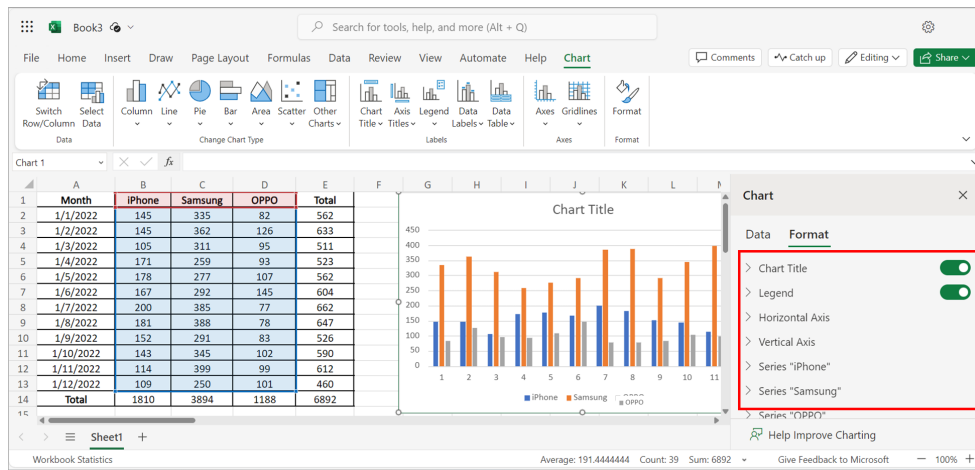
5. You can see the column chart in your Excel worksheet.



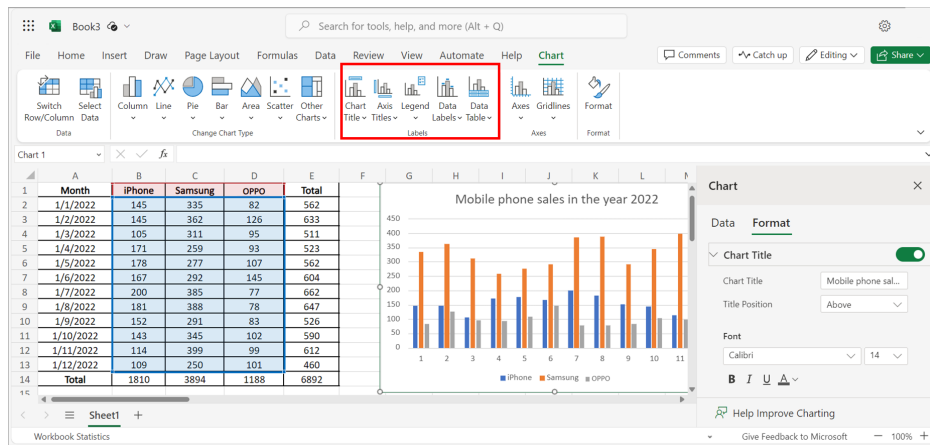
6. Select the **Chart** tab on the top ribbon and select **Format** option.



7. A **Chart** window will appear. You can view the option to format the chart including **Chart Title**, **Legend**, **Horizontal Axis**, **Vertical Axis**, and so on.



8. You can also format your column chart under the **Labels** section within **Chart** at the top ribbon.



Task 2: Create a Pie Chart

In this task, you'll learn to create a pie chart to represent the percentage sales for each mobile phone brand: iPhone, Samsung, and OPPO. Here, you'll compute the total sales for each mobile phone brand. To do so, follow the steps given below.

1. Select the **Total** sales of iPhone, Samsung, and OPPO in the Excel for the web worksheet.

Book3

Search for tools, help, and more (Alt + Q)

File Home Insert Draw Page Layout Formulas Data Review View Automate Help

Clipboard Font Alignment Number Styles Cells Editing Add-ins Analyze Data Show ToolPak

A14

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Month	iPhone	Samsung	OPPO	Total													
2	1/1/2022	145	335	82	562													
3	1/2/2022	145	362	126	633													
4	1/3/2022	105	311	95	511													
5	1/4/2022	171	259	93	523													
6	1/5/2022	178	277	107	562													
7	1/6/2022	167	292	145	604													
8	1/7/2022	200	385	77	662													
9	1/8/2022	181	388	78	647													
10	1/9/2022	152	291	83	526													
11	1/10/2022	143	345	102	590													
12	1/11/2022	114	399	99	612													
13	1/12/2022	109	250	101	460													
14	Total	1810	3894	1188	6892													

Workbook Statistics

Average: 2297.333333 Count: 4 Sum: 6892 Give Feedback to Microsoft 100%

2. Select the **Insert** tab and choose pie chart icon and then select the appropriate pie chart from it. In this case, let's choose the **2-D Pie** option.

Book3

Search for tools, help, and more (Alt + Q)

File Home **Insert** Draw Page Layout Formulas Data Review View Automate Help

PivotTable Table Forms Pictures Shapes Recommended Column Line Pie Bar Area Scatter Other Charts Hyperlink New Comment Text Box Text

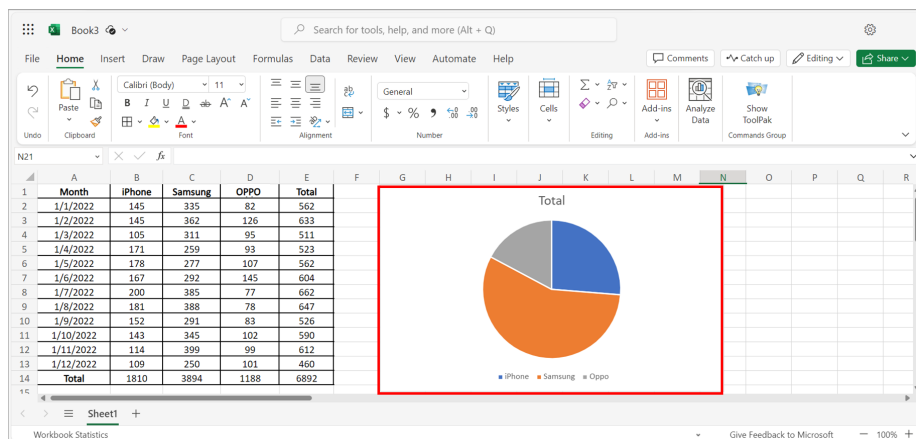
A14

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Month	iPhone	Samsung	OPPO	Total													
2	1/1/2022	145	335	82	562													
3	1/2/2022	145	362	126	633													
4	1/3/2022	105	311	95	511													
5	1/4/2022	171	259	93	523													
6	1/5/2022	178	277	107	562													
7	1/6/2022	167	292	145	604													
8	1/7/2022	200	385	77	662													
9	1/8/2022	181	388	78	647													
10	1/9/2022	152	291	83	526													
11	1/10/2022	143	345	102	590													
12	1/11/2022	114	399	99	612													
13	1/12/2022	109	250	101	460													
14	Total	1810	3894	1188	6892													

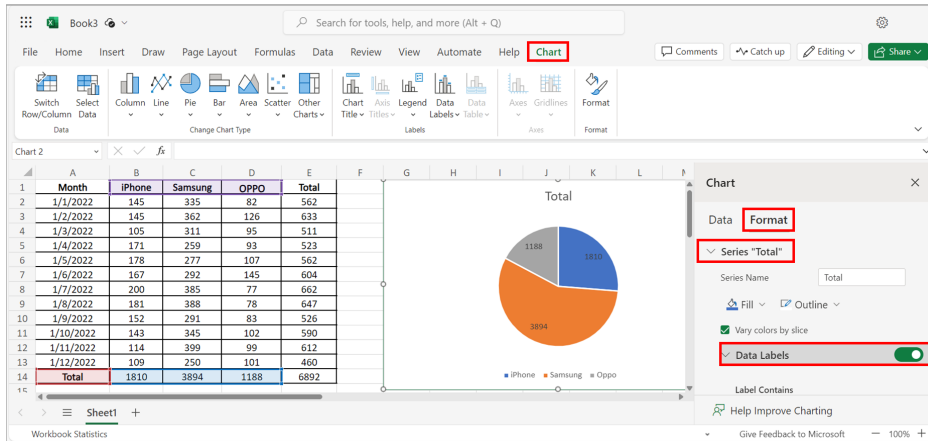
Workbook Statistics

Average: 2297.333333 Count: 4 Sum: 6892 Give Feedback to Microsoft 100%

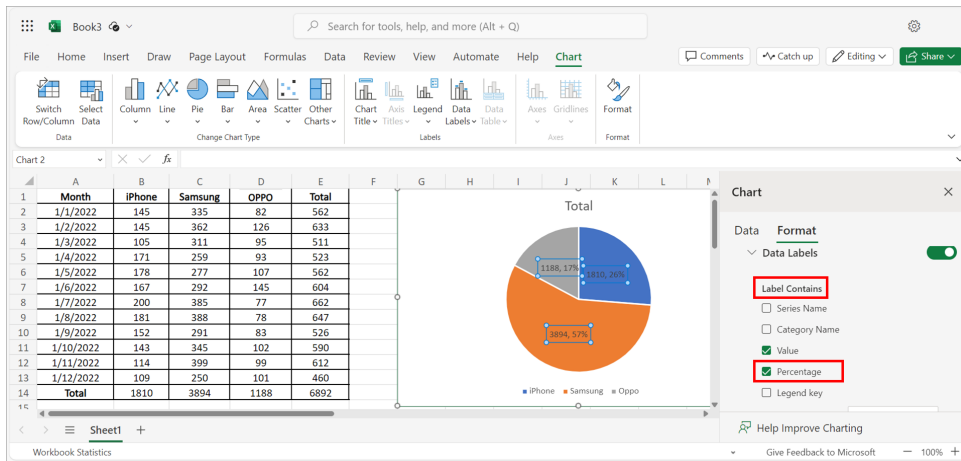
3. The Excel worksheet will show a pie chart for the given data set.



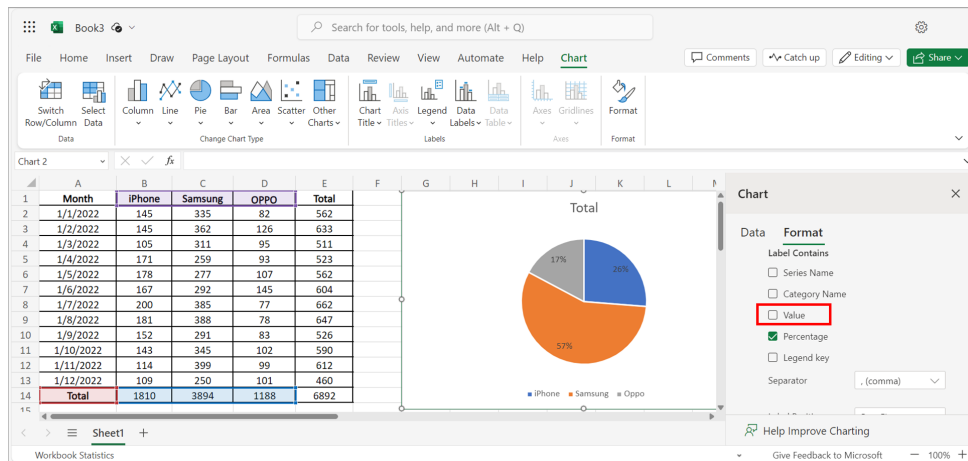
4. To present data in percentages, select **Chart** in the top ribbon. Then, select **Format** option from the **Chart** window. From the drop down list, toggle **Data Labels to Green**. You can see values in the pie chart.



5. Within **Label Contains**—select the **Percentage** checkbox. You'll see sales percentages for the three mobile phone brands.



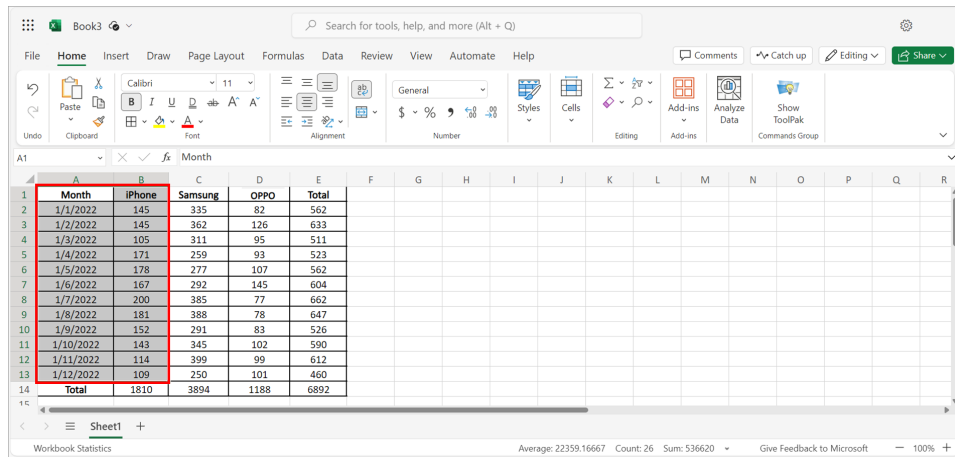
6. Uncheck the **Value** checkbox within the **Label Contains** to view only the sales percentage data.



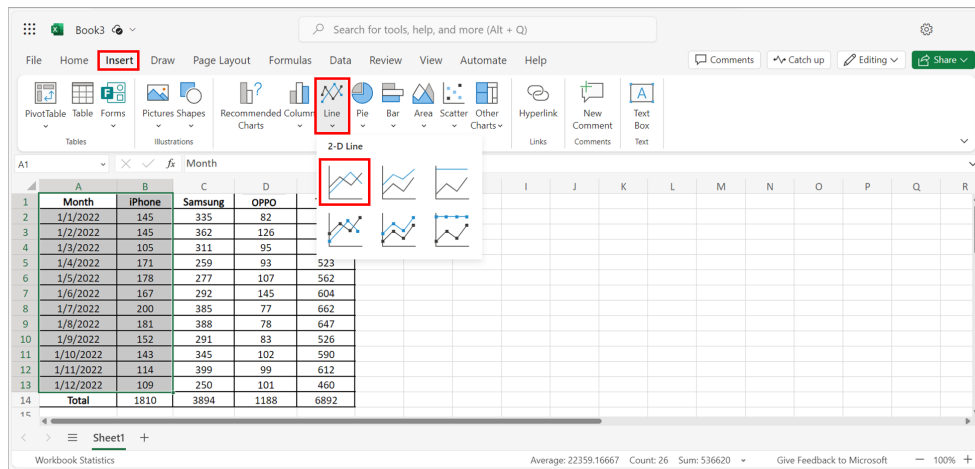
Task 3: Design a Line Chart

In this task, you'll learn to design the line chart. This chart helps identify trends and compare the data. To create a line chart, follow the below given steps.

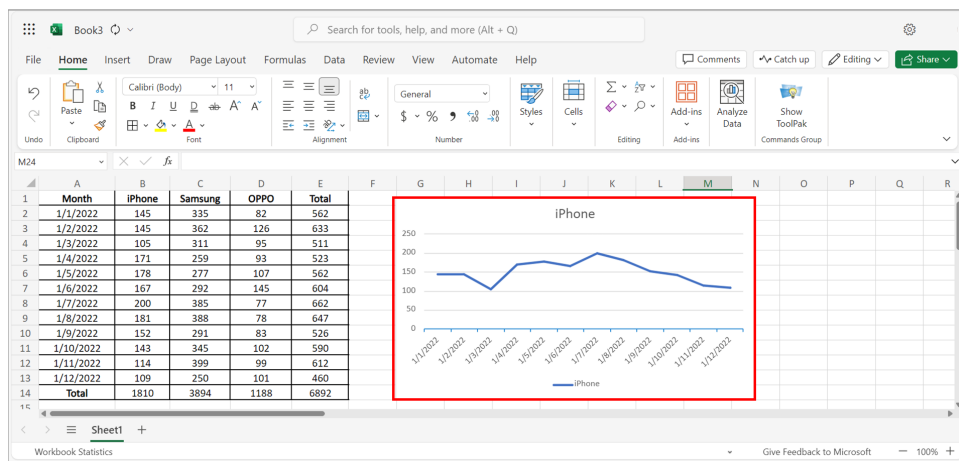
1. Select the data from the given data set. In this case, let's select monthly sales for iPhone for the year 2022.



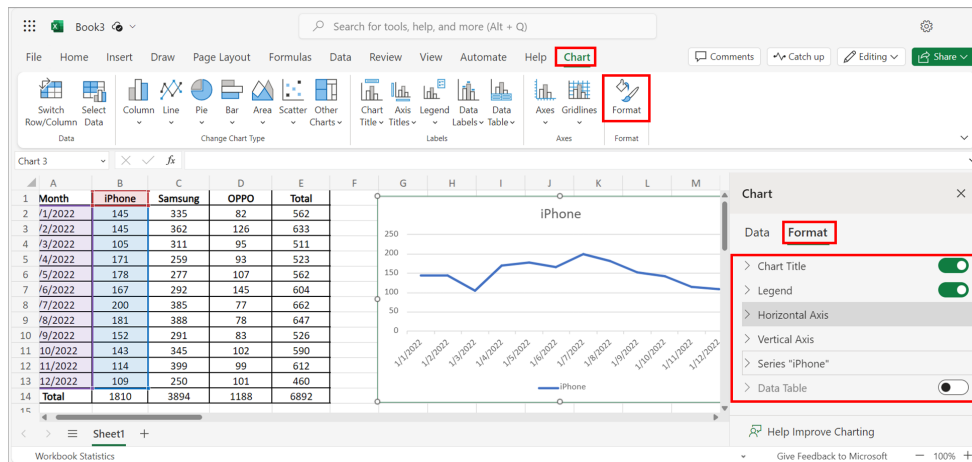
2. Select the **Insert** tab in the top ribbon and choose the appropriate line chart from the drop-down. In this case, let's select the **2-D Line** option.



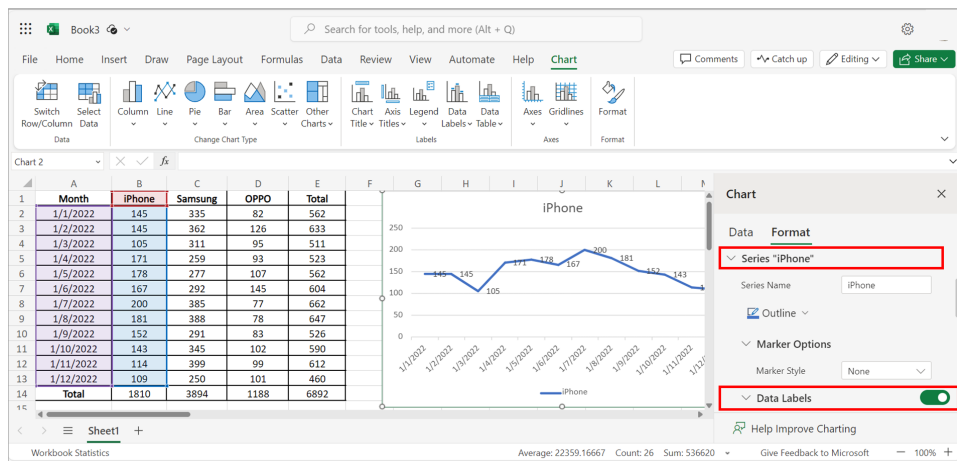
3. You'll see a line chart for the monthly sale of iPhones in the year 2022.



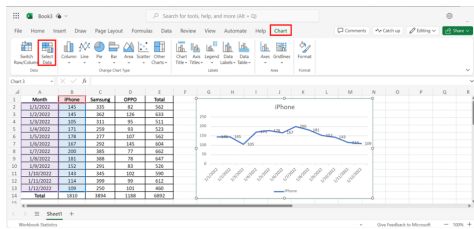
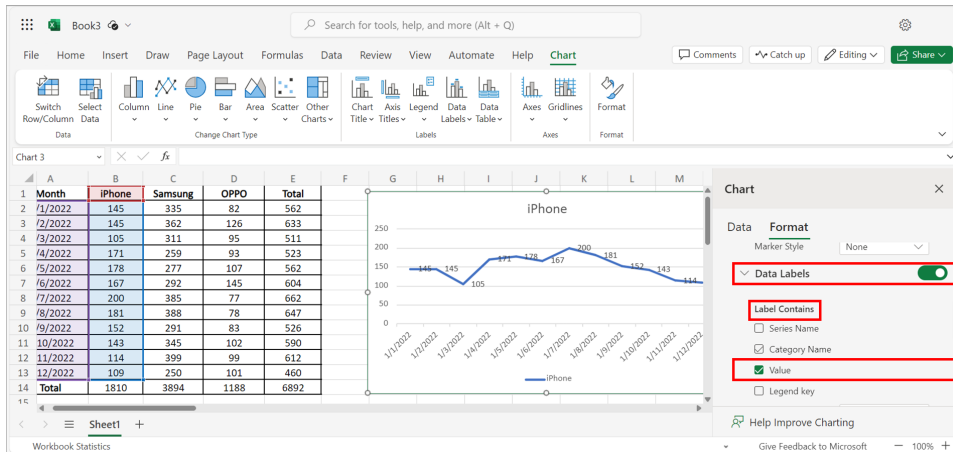
4. You can format the data by selecting the **Chart** tab at the top ribbon. Then, select the **Format** option. You can format **Chart Title**, **Legend**, **Horizontal Axis**, **Vertical Axis**, **Series iPhone**, and **Data Table**.



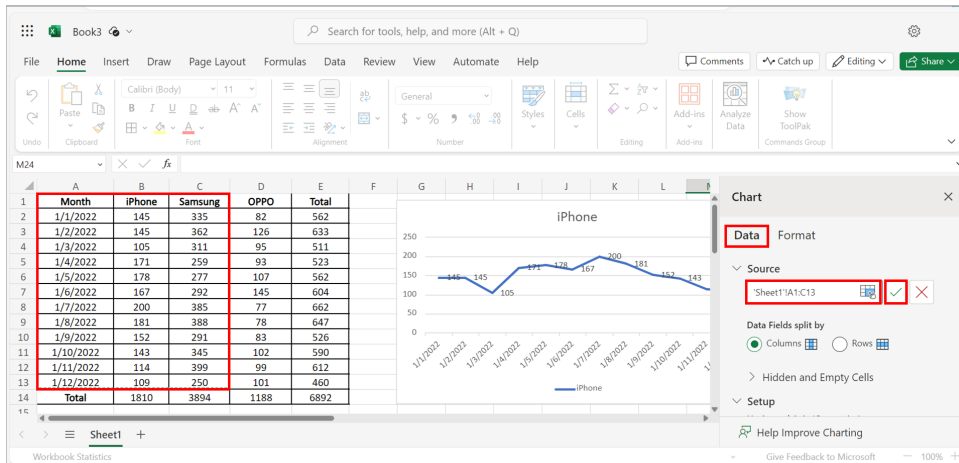
5. Select **Series iPhone**. A drop-down list will appear. Toggle **Data Label** to Green.



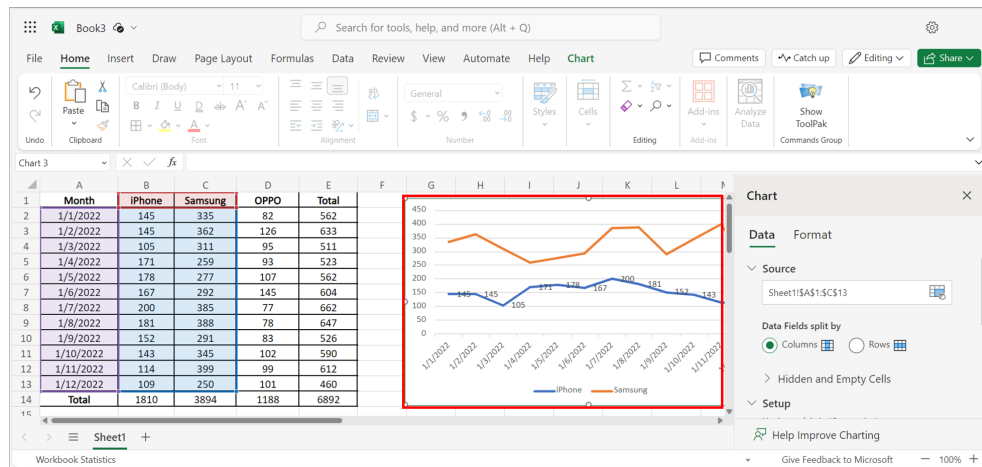
6. Within **Label Contains**, check or uncheck the **Values** checkbox to view or unview the data on the line chart.



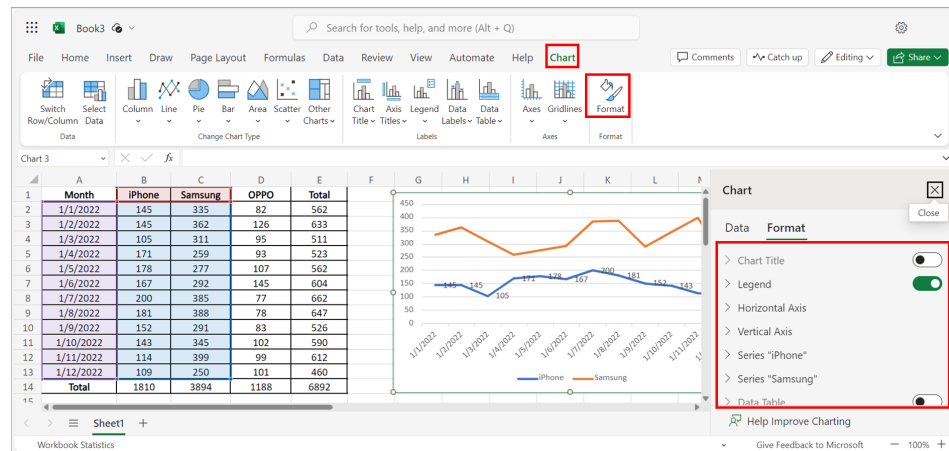
8. A **Chart** window will appear. Within the **Data** tab in the **Chart** window, select **Source** and choose the correct sign 'V'.



You'll see the comparison between the monthly sales data for iPhone and Samsung.



9. You can format this chart by selecting the **Format** option within the **Chart** tab at the top ribbon.

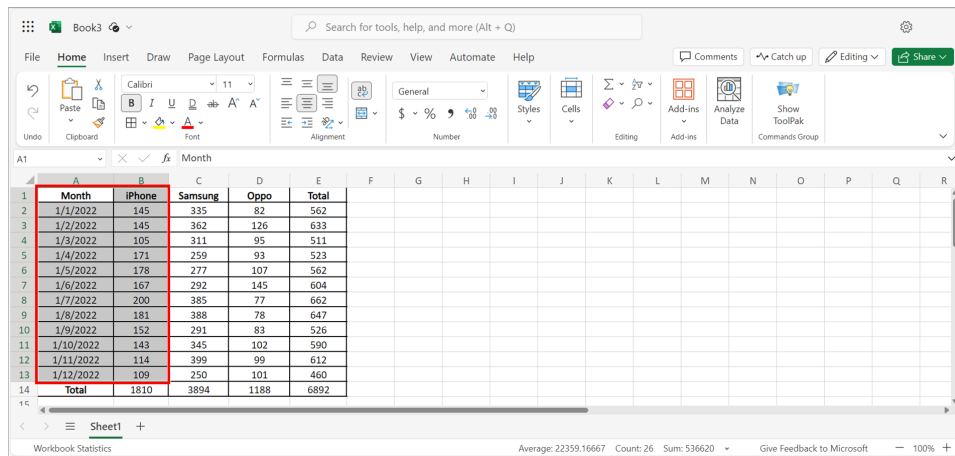


Task 4: Create a Waterfall Chart

In this task, you'll learn about the special type of chart. A waterfall chart helps to illustrate the contribution of the positive and negative values in total. It also helps to visualize changes in the data over time.

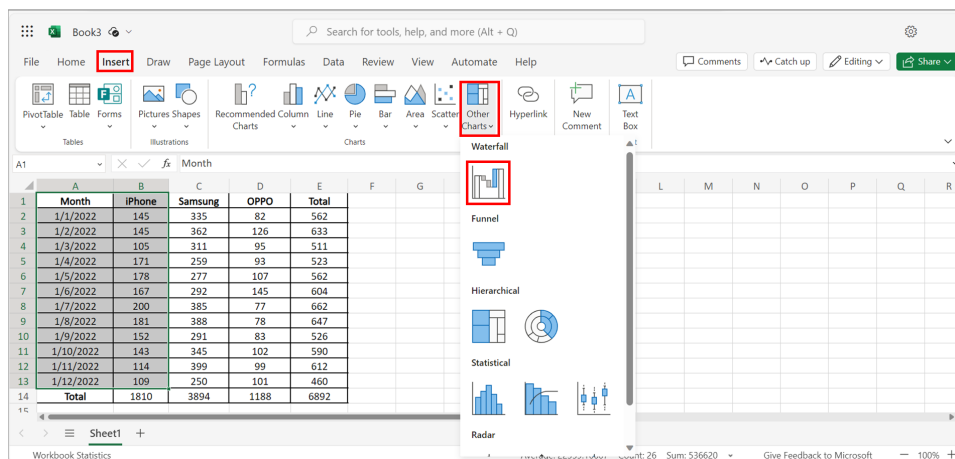
In this exercise, let's compare each month's sales for iPhones. To do so, follow the steps given below.

1. Select the data to create a waterfall chart.

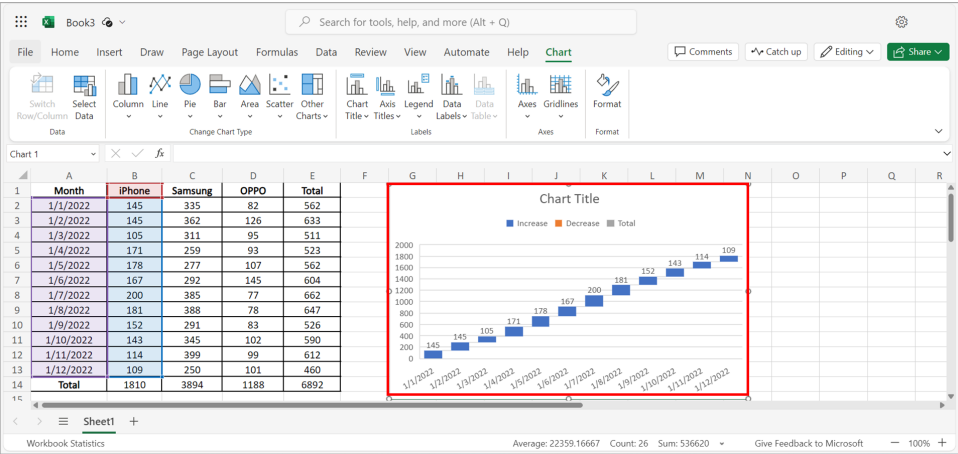


Month	iPhone	Samsung	Oppo	Total
1/1/2022	145	335	82	562
1/2/2022	145	362	126	633
1/3/2022	105	311	95	511
1/4/2022	171	259	93	523
1/5/2022	178	277	107	562
1/6/2022	167	292	145	604
1/7/2022	200	385	77	662
1/8/2022	181	388	78	647
1/9/2022	152	291	83	526
1/10/2022	143	345	102	590
1/11/2022	114	399	99	612
1/12/2022	109	250	101	460
Total	1810	3894	1188	6892

2. Select the **Insert** tab and then select the **Other Charts** option. Then, select the **Waterfall** chart from the options.



3. You'll see a chart showing monthly sales of iPhones for the given period.



There are many other visualization charts that you can use to present your data depending on the nature of the data and the objective of giving the information through a visualization chart.

Congratulations! You've completed lab Data Visualization Using Statistical Charts. You are now ready to move to the next topic.

Author(s)

- Dr. Rajendra Patel

Other Contributors

- Bhavika Chatur
- Rishi Kapoor

